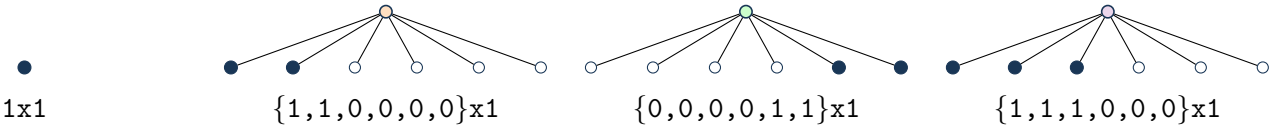


Definition and first terms

$7 A(x) = 6 + 1 x + A(x)^6$, with the origin-normalized solution.
Normalization/grammar: $A(x)=1+(1)*T(x)$; $T=x+15*T^2+20*T^3+15*T^4+6*T^5+1*T^6$.
 $a(0), \dots, a(7) = 1, 1, 15, 470, 18390, 805806, 37828981, 1860433080$.
Kernel used throughout: $\rho(u) = u(1 - 15 u^1 - 20 u^2 - 15 u^3 - 6 u^4 - 1 u^5)$, so $x = \rho(T(x))$.

Typogeometry



literal 6-slot geometry; empty positions are shown. Filled endpoints are true leaves; open endpoints are false/empty leaves. For colored grammars, color is genuine constructor data and is not silently converted into a positional zero pattern.

Typogeometric constructor codes and multiplicities: Δ_2 : 15, Δ_3 : 20, Δ_4 : 15, Δ_5 : 6, Δ_6 : 1.

Coefficient and contour extraction

$$\mathcal{K}_n = \{(k_1, \dots, k_5) \in \mathbb{Z}_{\geq 0}^5 : \sum_{j=1}^5 j k_j = n - 1\}, \quad K = \sum_{j=1}^5 k_j,$$
$$a(n) = \frac{1}{n} \sum_{\mathbf{k} \in \mathcal{K}_n} \binom{n + K - 1}{K} \binom{K}{k_1, \dots, k_5} 15^{k_1} 20^{k_2} 15^{k_3} 6^{k_4} 1^{k_5}.$$

$$a(n) = \frac{1}{2\pi i n} \oint_{\gamma} \frac{du}{\rho(u)^n}, \quad n \geq 1.$$

Recurrence

$\sum_{r=0}^5 P_r(n) a(n+r) = 0$ for $n \geq 1$. The exact degree-5 coefficient polynomials are embedded in `certificate.payload.json` (source: `data/recurrence.json`).

Differential equation

The scalar linear ODE has order 5. It is obtained from the recurrence by the standard Euler-operator substitution. Exact coefficients and boundary polynomial: `data/ode.json`.

Matrices, telescoper certificate, and checks

No compact matrix tuple is shown; see the embedded exact reduction data.

Certificate.

$$R(n, u) = N(n, u) / \rho(u)^4, \quad \deg_u N = 25.$$

$$\text{Telescoping identity: } \sum_r P_r(n) H_{n+r}(u) = \partial_u (R(n, u) H_n(u)).$$

Matrix dimensions: 12×12 ; remainder 5×6 , rank 5, nullity 1. The exact telescoping residual is zero. The recurrence reconstructs all 24 stored terms; 6/6 published OEIS prefix terms match.